Predicting the quality of red wine based on the chemical properties.

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving winequality-red.csv to winequality-red.csv
```

```
import pandas as pd
df = pd.read_csv("winequality-red.csv")
df.head()
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	5
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	5
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6

```
df.shape
```

```
(1599, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   fixed acidity         1599 non-null   float64
 1   volatile acidity      1599 non-null   float64
 2   citric acid           1599 non-null   float64
 3   residual sugar        1599 non-null   float64
 4   chlorides             1599 non-null   float64
 5   free sulfur dioxide   1599 non-null   float64
 6   total sulfur dioxide  1599 non-null   float64
 7   density               1599 non-null   float64
 8   pH                    1599 non-null   float64
 9   sulphates             1599 non-null   float64
 10  alcohol               1599 non-null   float64
 11  quality               1599 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB
```

```
df.describe()
```

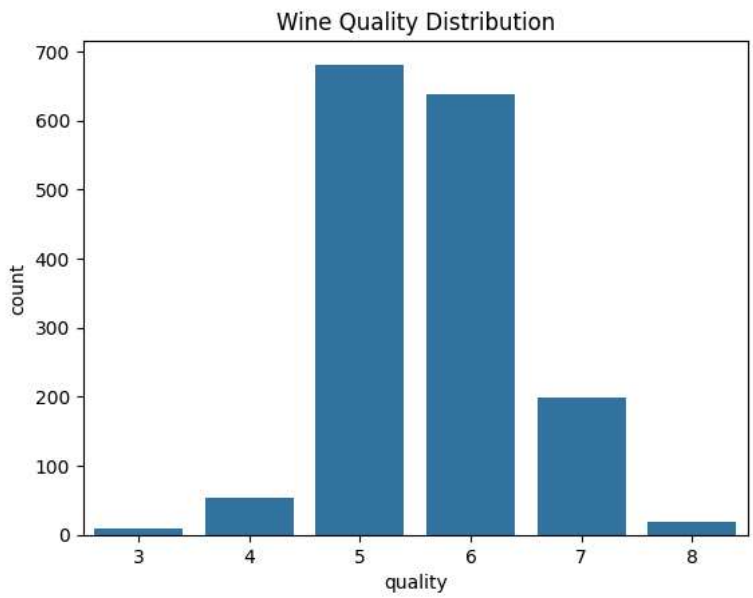
	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphat
count	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.0000
mean	8.319637	0.527821	0.270976	2.538806	0.087467	15.874922	46.467792	0.996747	3.311113	0.6581
std	1.741096	0.179060	0.194801	1.409928	0.047065	10.460157	32.895324	0.001887	0.154386	0.1695
min	4.600000	0.120000	0.000000	0.900000	0.012000	1.000000	6.000000	0.990070	2.740000	0.3300
25%	7.100000	0.390000	0.090000	1.900000	0.070000	7.000000	22.000000	0.995600	3.210000	0.5500
50%	7.900000	0.520000	0.260000	2.200000	0.079000	14.000000	38.000000	0.996750	3.310000	0.6200
75%	9.200000	0.640000	0.420000	2.600000	0.090000	21.000000	62.000000	0.997835	3.400000	0.7300

```
df.isnull().sum()
```

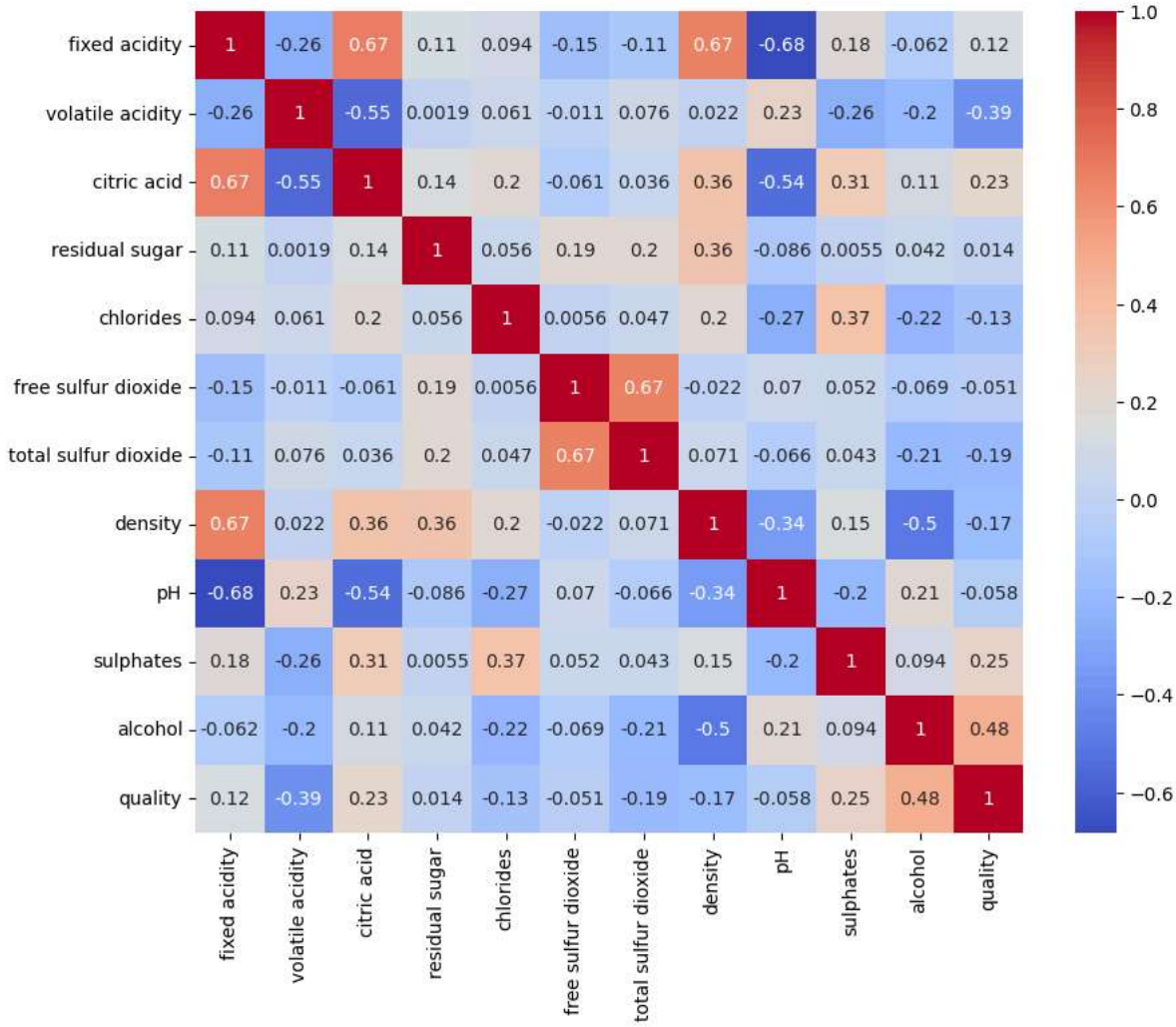
	0
fixed acidity	0
volatile acidity	0
citric acid	0
residual sugar	0
chlorides	0
free sulfur dioxide	0
total sulfur dioxide	0
density	0
pH	0
sulphates	0
alcohol	0
quality	0

dtype: int64

```
#quality distribution
import seaborn as sns
import matplotlib.pyplot as plt
sns.countplot(x='quality', data=df)
plt.title("Wine Quality Distribution")
plt.show()
```




```
#correlation heatmap
import seaborn as sns
import matplotlib.pyplot as plt
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.show()
```




```
#separate independent variables from target variables
X = df.drop("quality", axis=1)
y = df["quality"]
df.head()
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5
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3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6

```
print("X shape:", X.shape)
print("y shape:", y.shape)
```

```
X shape: (1599, 11)
y shape: (1599,)
```

```
#training and testing data
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

```
X_train shape: (1279, 11)
X_test shape: (320, 11)
y_train shape: (1279,)
y_test shape: (320,)
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
print(X_train.std())
```

```
1.0
```

```
print(X_train.mean())
```

```
2.298190858679605e-15
```

```
print(X_train.std())
```

```
1.0
```

```
print(X_train.shape)
print(X_test.shape)
```

```
(1279, 11)
(320, 11)
```

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
print(model.coef_)
```

```
[ 0.03978916 -0.17943891 -0.02751248  0.00942128 -0.08908432  0.05801491
 -0.12000836 -0.01920444 -0.06061037  0.14681503  0.2966275 ]
```

```
y_pred = model.predict(X_test)
print(y_pred[:10])
```

```
[5.34666441 5.05631345 5.66446972 5.46451484 5.72518476 5.27928659
 5.03421667 5.12623347 5.74534288 5.68665032]
```

```
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
print("R2 Score:", r2)
print("MAE:", mae)
print("MSE:", mse)
```

```
R2 Score: 0.40318034127906777
MAE: 0.5035304415524665
MSE: 0.3900251439643171
```

MAE (Mean Absolute Error)-Average difference between actual and predicted values. MSE (Mean Squared Error)-measures error R^2 Score-How well the model explains the data.